Absolutely! Let me walk through a complete step-by-step calculation for regression tree construction on a small dataset.

Dataset

Let's use this simple dataset:

X	y
1	15
2	20
3	25
4	30
5	35

Step 1: Calculate Root Node Statistics

All data in root node:

$y = [15, 20, 25, 30, 35]$

$\text{Mean} = (15 + 20 + 25 + 30 + 35) / 5 = 125 / 5 = 25$

$\text{MSE\_root} = \Sigma(y_i - \text{mean})^2 / n$
$= [(15-25)^2 + (20-25)^2 + (25-25)^2 + (30-25)^2 + (35-25)^2] / 5$
$= [100 + 25 + 0 + 25 + 100] / 5$
$= 250 / 5 = 50$

Initial MSE = 50

Step 2: Evaluate All Possible Splits

We consider splits between consecutive X values.

Split 1: X ≤ 1.5

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```text
Left node:  X ≤ 1.5 → [1] → y = [15]
Right node: X > 1.5 → [2,3,4,5] → y = [20,25,30,35]

Left node:
- Mean = 15
- MSE_left = (15-15)² = 0

Right node:
- Mean = (20+25+30+35)/4 = 110/4 = 27.5
- MSE_right = [(20-27.5)² + (25-27.5)² + (30-27.5)² + (35-27.5)²] / 4
  = [56.25 + 6.25 + 6.25 + 56.25] / 4
  = 125 / 4 = 31.25

Weighted MSE = (1×0 + 4×31.25) / 5 = 125/5 = 25
Gain = 50 - 25 = 25
```

Split 2: X ≤ 2.5

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```text
Left node:  X ≤ 2.5 → [1,2] → y = [15,20]
Right node: X > 2.5 → [3,4,5] → y = [25,30,35]

Left node:
- Mean = (15+20)/2 = 17.5
- MSE_left = [(15-17.5)² + (20-17.5)²] / 2 = [6.25 + 6.25]/2 = 12.5/2 = 6.25

Right node:
- Mean = (25+30+35)/3 = 90/3 = 30
- MSE_right = [(25-30)² + (30-30)² + (35-30)²] / 3 = [25 + 0 + 25]/3 = 50/3 ≈ 16.67

Weighted MSE = (2×6.25 + 3×16.67) / 5 = (12.5 + 50)/5 = 62.5/5 = 12.5
Gain = 50 - 12.5 = 37.5
```

Split 3: X ≤ 3.5

```text
Left node:  X ≤ 3.5 → [1,2,3] → y = [15,20,25]
Right node: X > 3.5 → [4,5] → y = [30,35]

Left node:
- Mean = (15+20+25)/3 = 60/3 = 20
- MSE_left = [(15-20)² + (20-20)² + (25-20)²] / 3 = [25 + 0 + 25]/3 = 50/3 ≈ 16.67

Right node:
- Mean = (30+35)/2 = 65/2 = 32.5
- MSE_right = [(30-32.5)² + (35-32.5)²] / 2 = [6.25 + 6.25]/2 = 12.5/2 = 6.25

Weighted MSE = (3×16.67 + 2×6.25) / 5 = (50 + 12.5)/5 = 62.5/5 = 12.5
Gain = 50 - 12.5 = 37.5
```

Split 4: X ≤ 4.5

```text
Left node:  X ≤ 4.5 → [1,2,3,4] → y = [15,20,25,30]
Right node: X > 4.5 → [5] → y = [35]

Left node:
- Mean = (15+20+25+30)/4 = 90/4 = 22.5
- MSE_left = [(15-22.5)² + (20-22.5)² + (25-22.5)² + (30-22.5)²] / 4
  = [56.25 + 6.25 + 6.25 + 56.25] / 4 = 125/4 = 31.25

Right node:
- Mean = 35
- MSE_right = (35-35)² = 0

Weighted MSE = (4×31.25 + 1×0) / 5 = 125/5 = 25
Gain = 50 - 25 = 25
```

Step 3: Choose Best Split

Split	Threshold	Weighted MSE	Gain
1	X ≤ 1.5	25	25
2	X ≤ 2.5	12.5	**37.5**
3	X ≤ 3.5	12.5	**37.5**
4	X ≤ 4.5	25	25

Best splits: X ≤ 2.5 or X ≤ 3.5 (both have gain = 37.5)

Let's choose **X ≤ 2.5** as our first split.

Step 4: Build Level 1 of Tree

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```text
Root: X ≤ 2.5?
    ├── Yes (Left): [1,2] → Mean = 17.5, MSE = 6.25
    └── No (Right): [3,4,5] → Mean = 30, MSE = 16.67
```

Step 5: Split Left Child (X ≤ 2.5 region)

Left Child Data: Points [1,2], y = [15,20], MSE = 6.25

Only one possible split: X ≤ 1.5

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```text
Left-Left:  X ≤ 1.5 → [1] → y = [15], MSE = 0
Left-Right: X > 1.5 → [2] → y = [20], MSE = 0

Weighted MSE = (1×0 + 1×0) / 2 = 0
Gain = 6.25 - 0 = 6.25
```

Since we have a perfect split (gain > 0), we'll take it.

Step 6: Split Right Child (X > 2.5 region)

Right Child Data: Points [3,4,5], y = [25,30,35], MSE = 16.67

Possible splits:

Split at X ≤ 3.5

```text
Right-Left:  X ≤ 3.5 → [3] → y = [25], MSE = 0
Right-Right: X > 3.5 → [4,5] → y = [30,35], MSE = 6.25

Weighted MSE = (1×0 + 2×6.25) / 3 = 12.5/3 ≈ 4.17
Gain = 16.67 - 4.17 = 12.5
```

Split at X ≤ 4.5

```text
Right-Left:  X ≤ 4.5 → [3,4] → y = [25,30], MSE = 6.25
Right-Right: X > 4.5 → [5] → y = [35], MSE = 0

Weighted MSE = (2×6.25 + 1×0) / 3 = 12.5/3 ≈ 4.17
Gain = 16.67 - 4.17 = 12.5
```

Both splits give the same gain. Let's choose **X ≤ 3.5**.

Step 7: Final Tree Structure

```text
Root: X ≤ 2.5?
    ├── Yes → X ≤ 1.5?
    │   ├── Yes: Predict 15 (Leaf)
    │   └── No: Predict 20 (Leaf)
    └── No → X ≤ 3.5?
        ├── Yes: Predict 25 (Leaf)
        └── No: Predict 32.5 (Leaf)
```

Step 8: Verify Predictions

Let's test our tree:

- **X = 1**: X ≤ 2.5? Yes → X ≤ 1.5? Yes → Predict 15 ✓
- **X = 2**: X ≤ 2.5? Yes → X ≤ 1.5? No → Predict 20 ✓
- **X = 3**: X ≤ 2.5? No → X ≤ 3.5? Yes → Predict 25 ✓
- **X = 4**: X ≤ 2.5? No → X ≤ 3.5? No → Predict 32.5 ✓
- **X = 5**: X ≤ 2.5? No → X ≤ 3.5? No → Predict 32.5 ✓

Step 9: Calculate Final MSE

Actual: [15, 20, 25, 30, 35]
Predicted: [15, 20, 25, 32.5, 32.5]

$MSE = [(15-15)^2 + (20-20)^2 + (25-25)^2 + (30-32.5)^2 + (35-32.5)^2] / 5$
$= [0 + 0 + 0 + 6.25 + 6.25] / 5 = 12.5/5 = 2.5$

We reduced MSE from 50 to 2.5!

Summary of Key Calculations

- **Initial MSE**: 50
- **Best first split**: X ≤ 2.5 (gain = 37.5)
- **Left child split**: X ≤ 1.5 (gain = 6.25)
- **Right child split**: X ≤ 3.5 (gain = 12.5)
- **Final MSE**: 2.5

This shows how regression trees recursively partition the data to minimize prediction error at each step!